

data_preview

May 14, 2026

1 HW3 Data Preview

1.1 Head of each data source

NASDAQ Composite (raw daily) | Series | shape=(4266,)

NASDAQ Composite (raw daily)	
2009-06-01	1828.68
2009-06-02	1836.80
2009-06-03	1825.92
2009-06-04	1850.02
2009-06-05	1849.42

Euro Area share-price index (raw monthly) | Series | shape=(200,)

Euro Area share-price index (raw monthly)	
2009-06-01	63.63773
2009-07-01	64.00575
2009-08-01	70.37932
2009-09-01	74.09707
2009-10-01	75.43439

UK share-price index (raw monthly) | Series | shape=(202,)

UK share-price index (raw monthly)	
2009-06-01	65.94963
2009-07-01	66.33266
2009-08-01	72.22261
2009-09-01	76.31963
2009-10-01	78.26147

Japan Nikkei 225 (raw daily) | Series | shape=(4145,)

Japan Nikkei 225 (raw daily)	
2009-06-01	9677.75
2009-06-02	9704.31
2009-06-03	9741.67
2009-06-04	9668.96
2009-06-05	9768.01

China share-price index (raw monthly) | Series | shape=(202,)

China share-price index (raw monthly)	
2009-06-01	76.57807
2009-07-01	86.84656
2009-08-01	83.20996
2009-09-01	78.16435
2009-10-01	79.71270

Canada share-price index (raw monthly) | Series | shape=(202,)

Canada share-price index (raw monthly)	
2009-06-01	72.33577
2009-07-01	71.80637
2009-08-01	75.42292
2009-09-01	78.41676
2009-10-01	78.79231

India share-price index (raw monthly) | Series | shape=(202,)

India share-price index (raw monthly)	
2009-06-01	54.03560
2009-07-01	53.49723
2009-08-01	56.34653
2009-09-01	59.84916
2009-10-01	61.58843

Brazil share-price index (raw monthly) | Series | shape=(202,)

Brazil share-price index (raw monthly)	
2009-06-01	104.7077
2009-07-01	104.3460
2009-08-01	113.8281
2009-09-01	118.6840
2009-10-01	128.5557

US 10Y Treasury yield (raw daily, %) | Series | shape=(4240,)

US 10Y Treasury yield (raw daily, %)	
2009-06-01	3.71
2009-06-02	3.65
2009-06-03	3.56
2009-06-04	3.72
2009-06-05	3.84

Broad USD index (raw daily) | Series | shape=(4218,)

Broad USD index (raw daily)	
2009-06-01	94.8107
2009-06-02	94.6556
2009-06-03	95.0751
2009-06-04	95.2922
2009-06-05	96.0412

WTI Crude Oil (raw daily) | Series | shape=(4252,)

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                WTI Crude Oil (raw daily)
2009-06-01      68.59
2009-06-02      68.58
2009-06-03      66.14
2009-06-04      68.80
2009-06-05      68.43

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Global all-commodities index (raw monthly) | Series | shape=(202,)

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                Global all-commodities index (raw monthly)
2009-06-01      120.511184
2009-07-01      116.330294
2009-08-01      122.804676
2009-09-01      119.114792
2009-10-01      126.625529

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Monthly level panel (12 assets) | DataFrame | shape=(204, 12)

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                US_NASDAQ  EuroArea_Eq    UK_Eq  Japan_Nikkei  China_Eq  \
2009-06-30      1835.04      63.63773  65.94963      9958.44  76.57807
2009-07-31      1978.50      64.00575  66.33266      10356.83  86.84656
2009-08-31      2009.06      70.37932  72.22261      10492.53  83.20996
2009-09-30      2122.42      74.09707  76.31963      10133.23  78.16435
2009-10-31      2045.11      75.43439  78.26147      10034.74  79.71270

                Canada_Eq  India_Eq  Brazil_Eq  US_10Y_TR  USD_Broad  WTI_Oil  \
2009-06-30      72.33577  54.03560  104.7077      3.53      96.3761  69.82
2009-07-31      71.80637  53.49723  104.3460      3.52      94.2526  69.26
2009-08-31      75.42292  56.34653  113.8281      3.40      94.6202  69.97
2009-09-30      78.41676  59.84916  118.6840      3.31      93.1907  70.46
2009-10-31      78.79231  61.58843  128.5557      3.41      92.6207  77.04

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                GlobalCommod
2009-06-30      120.511184
2009-07-31      116.330294
2009-08-31      122.804676
2009-09-30      119.114792
2009-10-31      126.625529

```

Monthly returns (12 assets) | DataFrame | shape=(193, 12)

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                US_NASDAQ  EuroArea_Eq    UK_Eq  Japan_Nikkei  China_Eq  \
2010-01-31     -0.055171      0.012328  0.017543     -0.033593 -0.015975
2010-02-28      0.041464     -0.062406 -0.033787     -0.007086 -0.056585
2010-03-31      0.068920      0.059005  0.071735      0.090929  0.020501
2010-04-30      0.026027      0.022543  0.017968     -0.002939  0.000428
2010-05-31     -0.086591     -0.098537 -0.088738     -0.123916 -0.135053

                Canada_Eq  India_Eq  Brazil_Eq  US_10Y_TR  USD_Broad  WTI_Oil  \
2010-01-31      0.005450  0.007769  0.005673  0.021908  0.009122 -0.085970
2010-02-28     -0.017667 -0.062912 -0.037760  0.004725 -0.001106  0.090118

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2010-03-31	0.043750	0.064298	0.046543	-0.016542	-0.008164	0.045727
2010-04-30	0.015760	0.023721	0.010258	0.015950	-0.000607	0.030913
2010-05-31	-0.029586	-0.048736	-0.108533	0.035375	0.036630	-0.151096

	GlobalCommod
2010-01-31	0.042102
2010-02-28	-0.023711
2010-03-31	0.035602
2010-04-30	0.053689
2010-05-31	-0.061251

Full-sample monthly std (per asset) | Series | shape=(12,)

	Full-sample monthly std (per asset)
US_NASDAQ	0.049181
EuroArea_Eq	0.041605
UK_Eq	0.032700
Japan_Nikkei	0.049703
China_Eq	0.049104

Standardized monthly returns (units of std) | DataFrame | shape=(193, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2010-01-31	-1.121790	0.296318	0.536477	-0.675877	-0.325330	
2010-02-28	0.843094	-1.499975	-1.033224	-0.142572	-1.152341	
2010-03-31	1.401345	1.418217	2.193705	1.829464	0.417494	
2010-04-30	0.529200	0.541827	0.549467	-0.059122	0.008712	
2010-05-31	-1.760653	-2.368412	-2.713668	-2.493161	-2.750325	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2010-01-31	0.174359	0.205647	0.106040	1.092524	0.552081	-0.761720	
2010-02-28	-0.565195	-1.665237	-0.705867	0.235626	-0.066910	0.798473	
2010-03-31	1.399599	1.701906	0.870061	-0.824899	-0.494096	0.405157	
2010-04-30	0.504171	0.627881	0.191763	0.795394	-0.036756	0.273901	
2010-05-31	-0.946492	-1.289992	-2.028865	1.764080	2.217002	-1.338756	

	GlobalCommod
2010-01-31	1.000534
2010-02-28	-0.563463
2010-03-31	0.846067
2010-04-30	1.275876
2010-05-31	-1.455586

Cumulative gross returns (Growth of \$1) | DataFrame | shape=(193, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2010-01-31	0.944829	1.012328	1.017543	0.966407	0.984025	
2010-02-28	0.984006	0.949153	0.983163	0.959559	0.928344	
2010-03-31	1.051823	1.005157	1.053691	1.046811	0.947376	
2010-04-30	1.079199	1.027816	1.072623	1.043735	0.947781	
2010-05-31	0.985750	0.926538	0.977441	0.914399	0.819780	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2010-01-31	1.005450	1.007769	1.005673	1.021908	1.009122	0.914030	
2010-02-28	0.987687	0.944368	0.967698	1.026737	1.008006	0.996401	
2010-03-31	1.030898	1.005089	1.012738	1.009753	0.999777	1.041963	
2010-04-30	1.047145	1.028931	1.023127	1.025858	0.999170	1.074174	
2010-05-31	1.016164	0.978785	0.912084	1.062148	1.035770	0.911871	

	GlobalCommod
2010-01-31	1.042102
2010-02-28	1.017394
2010-03-31	1.053615
2010-04-30	1.110183
2010-05-31	1.042183

Industrial Production INDPRO (raw) | Series | shape=(219,)

	Industrial Production INDPRO (raw)
2008-01-01	102.2448
2008-02-01	101.8438
2008-03-01	101.5336
2008-04-01	100.8785
2008-05-01	100.2674

CPI Headline CPIAUCSL (raw) | Series | shape=(219,)

	CPI Headline CPIAUCSL (raw)
2008-01-01	212.174
2008-02-01	212.687
2008-03-01	213.448
2008-04-01	213.942
2008-05-01	215.208

VIX daily VIXCLS (raw) | Series | shape=(4648,)

	VIX daily VIXCLS (raw)
2008-01-02	23.17
2008-01-03	22.49
2008-01-04	23.94
2008-01-07	23.79
2008-01-08	25.43

INDPRO 12m log-change (IP YoY) | Series | shape=(219,)

	INDPRO 12m log-change (IP YoY)
2009-01-01	-0.143577
2009-02-01	-0.145751
2009-03-01	-0.158480
2009-04-01	-0.160333
2009-05-01	-0.164407

CPI 12m log-change (CPI YoY) | Series | shape=(219,)

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CPI 12m log-change (CPI YoY)
2009-01-01      -0.001137
2009-02-01       0.000085
2009-03-01      -0.004475
2009-04-01      -0.005780
2009-05-01      -0.010210

VIX monthly average | Series | shape=(221,)

VIX monthly average
2008-01-31      25.816190
2008-02-29      25.456000
2008-03-31      27.103500
2008-04-30      21.562727
2008-05-31      18.303333

dVIX (first diff of monthly VIX) | Series | shape=(221,)

dVIX (first diff of monthly VIX)
2008-02-29      -0.360190
2008-03-31       1.647500
2008-04-30      -5.540773
2008-05-31      -3.259394
2008-06-30       3.807143

Macro factor panel (IP_YoY, CPI_YoY, dVIX) | DataFrame | shape=(194, 3)

IP_YoY  CPI_YoY  dVIX
2010-01-31  0.008685  0.025873  -0.596388
2010-02-28  0.018535  0.021285   1.896842
2010-03-31  0.041148  0.022604  -4.772609
2010-04-30  0.052860  0.021828  -0.343106
2010-05-31  0.076430  0.019837  14.505214

In-sample joint panel (returns + macro) | DataFrame | shape=(180, 15)

US_NASDAQ  EuroArea_Eq  UK_Eq  Japan_Nikkei  China_Eq  \
2010-01-31  -1.121790    0.296318  0.536477   -0.675877  -0.325330
2010-02-28   0.843094   -1.499975 -1.033224   -0.142572  -1.152341
2010-03-31   1.401345    1.418217  2.193705    1.829464   0.417494
2010-04-30   0.529200    0.541827  0.549467   -0.059122   0.008712
2010-05-31  -1.760653   -2.368412 -2.713668   -2.493161  -2.750325

Canada_Eq  India_Eq  Brazil_Eq  US_10Y_TR  USD_Broad  WTI_Oil  \
2010-01-31   0.174359  0.205647   0.106040   1.092524   0.552081  -0.761720
2010-02-28  -0.565195 -1.665237  -0.705867   0.235626  -0.066910  0.798473
2010-03-31   1.399599  1.701906   0.870061  -0.824899  -0.494096  0.405157
2010-04-30   0.504171  0.627881   0.191763   0.795394  -0.036756  0.273901
2010-05-31  -0.946492 -1.289992  -2.028865   1.764080   2.217002 -1.338756

GlobalCommod  IP_YoY  CPI_YoY  dVIX
2010-01-31    1.000534  0.008685  0.025873  -0.596388

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2010-02-28	-0.563463	0.018535	0.021285	1.896842
2010-03-31	0.846067	0.041148	0.022604	-4.772609
2010-04-30	1.275876	0.052860	0.021828	-0.343106
2010-05-31	-1.455586	0.076430	0.019837	14.505214

Macro regression coefficient/ R^2 table | DataFrame | shape=(12, 7)

	b_IP	t_IP	b_CPI	t_CPI	b_dVIX	t_dVIX	\
asset							
US_NASDAQ	-0.982005	-0.661050	-7.710366	-2.113955	-0.122195	-5.232166	
EuroArea_Eq	2.205022	1.308014	-4.392199	-1.299574	-0.167337	-11.935488	
UK_Eq	3.228904	1.819837	0.282183	0.116612	-0.177775	-17.386835	
Japan_Nikkei	-1.399712	-0.918828	-3.080146	-1.207452	-0.107679	-4.984310	
China_Eq	-0.343664	-0.207295	-5.316004	-1.138076	-0.048575	-2.184786	

	R2
asset	
US_NASDAQ	0.343654
EuroArea_Eq	0.574699
UK_Eq	0.636124
Japan_Nikkei	0.263703
China_Eq	0.060027

OOS joint panel (2025+) | DataFrame | shape=(12, 15)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2025-01-31	0.330708	0.720444	0.631568	-0.163076	-0.943411	
2025-02-28	-0.824708	1.404374	1.118132	-1.267990	0.568464	
2025-03-31	-1.742606	0.230466	-0.177897	-0.850520	0.286737	
2025-04-30	0.172107	-1.723973	-1.513939	0.240228	-0.630162	
2025-05-31	1.855982	1.873857	1.500409	1.043983	0.480375	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2025-01-31	-0.032008	-0.998806	-0.296870	0.190329	-0.373202	0.048790	
2025-02-28	0.412595	-0.356254	0.626255	1.631514	-0.095371	-0.356173	
2025-03-31	-0.808916	-0.267343	0.335771	0.218588	-0.746740	0.237389	
2025-04-30	-0.885491	0.659403	0.283283	0.430111	-1.998412	-1.666116	
2025-05-31	1.993128	1.346431	0.987584	-0.844015	-0.408530	0.279722	

	GlobalCommod	IP_YoY	CPI_YoY	dVIX
2025-01-31	0.866294	0.008454	0.029471	0.897446
2025-02-28	-0.083186	0.008087	0.027631	0.204364
2025-03-31	-0.648017	0.005786	0.023541	4.873429
2025-04-30	-0.704249	0.008785	0.022988	10.124762
2025-05-31	-0.228456	0.001016	0.023494	-11.503918

OOS macro predictions | DataFrame | shape=(12, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2025-01-31	0.097544	-0.091567	-0.092694	0.037459	-0.060985	
2025-02-28	0.196788	0.031686	0.028815	0.118273	-0.017407	

2025-03-31	-0.339955	-0.736731	-0.809815	-0.368668	-0.221672
2025-04-30	-0.980326	-1.606430	-1.733845	-0.936621	-0.474846
2025-05-31	1.666322	1.993486	2.086259	1.401646	0.575737

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil \
2025-01-31	-0.030947	0.102114	-0.057244	0.082119	0.204122	-0.108032
2025-02-28	0.089451	0.190746	0.028069	0.071366	0.139472	-0.027710
2025-03-31	-0.686490	-0.390829	-0.518586	0.278785	0.534233	-0.596643
2025-04-30	-1.559329	-1.038185	-1.135060	0.488578	0.982844	-1.232166
2025-05-31	2.049208	1.641926	1.413079	-0.349886	-0.875443	1.379527

	GlobalCommod
2025-01-31	-0.037698
2025-02-28	0.013538
2025-03-31	-0.391889
2025-04-30	-0.823951
2025-05-31	0.959654

OOS macro residuals (asset - predicted) | DataFrame | shape=(12, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq \
2025-01-31	0.233164	0.812011	0.724261	-0.200535	-0.882426
2025-02-28	-1.021496	1.372688	1.089317	-1.386263	0.585871
2025-03-31	-1.402651	0.967197	0.631919	-0.481852	0.508409
2025-04-30	1.152433	-0.117543	0.219906	1.176849	-0.155316
2025-05-31	0.189661	-0.119630	-0.585851	-0.357664	-0.095362

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil \
2025-01-31	-0.001061	-1.100920	-0.239625	0.108210	-0.577323	0.156823
2025-02-28	0.323145	-0.547001	0.598186	1.560148	-0.234843	-0.328463
2025-03-31	-0.122426	0.123486	0.854357	-0.060197	-1.280973	0.834031
2025-04-30	0.673838	1.697588	1.418344	-0.058467	-2.981256	-0.433950
2025-05-31	-0.056080	-0.295495	-0.425495	-0.494130	0.466913	-1.099805

	GlobalCommod
2025-01-31	0.903992
2025-02-28	-0.096724
2025-03-31	-0.256128
2025-04-30	0.119702
2025-05-31	-1.188109

OOS macro lead/lag summary | DataFrame | shape=(12, 3)

	cum_actual_std	cum_pred_std	cum_resid_std
Canada_Eq	7.464940	1.900202	5.564738
Brazil_Eq	5.757664	0.915054	4.842610
UK_Eq	5.407304	1.248959	4.158344
China_Eq	3.504925	0.022335	3.482591
EuroArea_Eq	4.238217	1.225002	3.013215

In-sample standardized panel (PCA input) | DataFrame | shape=(180, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2010-01-31	-1.121790	0.296318	0.536477	-0.675877	-0.325330	
2010-02-28	0.843094	-1.499975	-1.033224	-0.142572	-1.152341	
2010-03-31	1.401345	1.418217	2.193705	1.829464	0.417494	
2010-04-30	0.529200	0.541827	0.549467	-0.059122	0.008712	
2010-05-31	-1.760653	-2.368412	-2.713668	-2.493161	-2.750325	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2010-01-31	0.174359	0.205647	0.106040	1.092524	0.552081	-0.761720	
2010-02-28	-0.565195	-1.665237	-0.705867	0.235626	-0.066910	0.798473	
2010-03-31	1.399599	1.701906	0.870061	-0.824899	-0.494096	0.405157	
2010-04-30	0.504171	0.627881	0.191763	0.795394	-0.036756	0.273901	
2010-05-31	-0.946492	-1.289992	-2.028865	1.764080	2.217002	-1.338756	

	GlobalCommod
2010-01-31	1.000534
2010-02-28	-0.563463
2010-03-31	0.846067
2010-04-30	1.275876
2010-05-31	-1.455586

PCA loadings PC1-PC3 | DataFrame | shape=(12, 3)

	PC1	PC2	PC3
US_NASDAQ	0.269940	0.495075	0.208863
EuroArea_Eq	0.362554	0.048097	0.067898
UK_Eq	0.370509	-0.038857	-0.048551
Japan_Nikkei	0.258021	0.208951	0.533913
China_Eq	0.171830	-0.035948	0.458927

In-sample PC scores (PC1-PC3) | DataFrame | shape=(180, 3)

	PC1	PC2	PC3
2010-01-31	-0.575452	-0.630272	-1.421741
2010-02-28	-2.097574	0.486172	-0.116189
2010-03-31	3.873681	0.115836	0.703737
2010-04-30	1.016949	0.030903	-0.944745
2010-05-31	-6.658297	-0.315068	-1.905571

OOS PC scores (projected) | DataFrame | shape=(13, 3)

	PC1	PC2	PC3
2025-01-31	0.261340	0.091300	-0.660861
2025-02-28	0.374710	0.367523	-1.355509
2025-03-31	-0.856827	-0.387064	-0.700706
2025-04-30	-1.479768	1.882249	-0.456880
2025-05-31	3.787607	0.694830	0.839715

OOS PC predictions | DataFrame | shape=(13, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2025-01-31	0.219596	0.138265	0.199528	-0.117621	-0.255887	
2025-02-28	0.241864	0.145488	0.264525	-0.401534	-0.565129	
2025-03-31	-0.327390	-0.292844	-0.194240	-0.527360	-0.449111	
2025-04-30	0.678860	-0.392992	-0.525063	-0.083735	-0.525829	
2025-05-31	1.783685	1.547642	1.409737	1.719514	1.016989	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2025-01-31	0.318653	0.412798	0.274970	0.397334	-0.151904	0.142070	
2025-02-28	0.438467	0.561729	0.437976	0.853284	-0.463279	0.209099	
2025-03-31	-0.052876	0.048971	-0.040139	0.287843	0.290454	-0.074807	
2025-04-30	-0.461500	-0.123047	-0.413909	1.602530	-0.416091	-0.791108	
2025-05-31	1.358344	1.326909	1.036639	-0.401971	-0.838727	0.838149	

	GlobalCommod
2025-01-31	0.262142
2025-02-28	0.397173
2025-03-31	0.157306
2025-04-30	-0.931557
2025-05-31	0.519235

OOS PC residuals | DataFrame | shape=(13, 12)

	US_NASDAQ	EuroArea_Eq	UK_Eq	Japan_Nikkei	China_Eq	\
2025-01-31	0.111112	0.582179	0.432039	-0.045455	-0.687524	
2025-02-28	-1.066572	1.258886	0.853607	-0.866456	1.133592	
2025-03-31	-1.415216	0.523310	0.016344	-0.323160	0.735847	
2025-04-30	-0.506752	-1.330980	-0.988876	0.323962	-0.104333	
2025-05-31	0.072297	0.326214	0.090672	-0.675531	-0.536614	

	Canada_Eq	India_Eq	Brazil_Eq	US_10Y_TR	USD_Broad	WTI_Oil	\
2025-01-31	-0.350661	-1.411604	-0.571839	-0.207005	-0.221298	-0.093280	
2025-02-28	-0.025872	-0.917983	0.188279	0.778230	0.367908	-0.565271	
2025-03-31	-0.756040	-0.316314	0.375910	-0.069256	-1.037194	0.312195	
2025-04-30	-0.423991	0.782450	0.697193	-1.172419	-1.582321	-0.875007	
2025-05-31	0.634784	0.019522	-0.049055	-0.442045	0.430197	-0.558427	

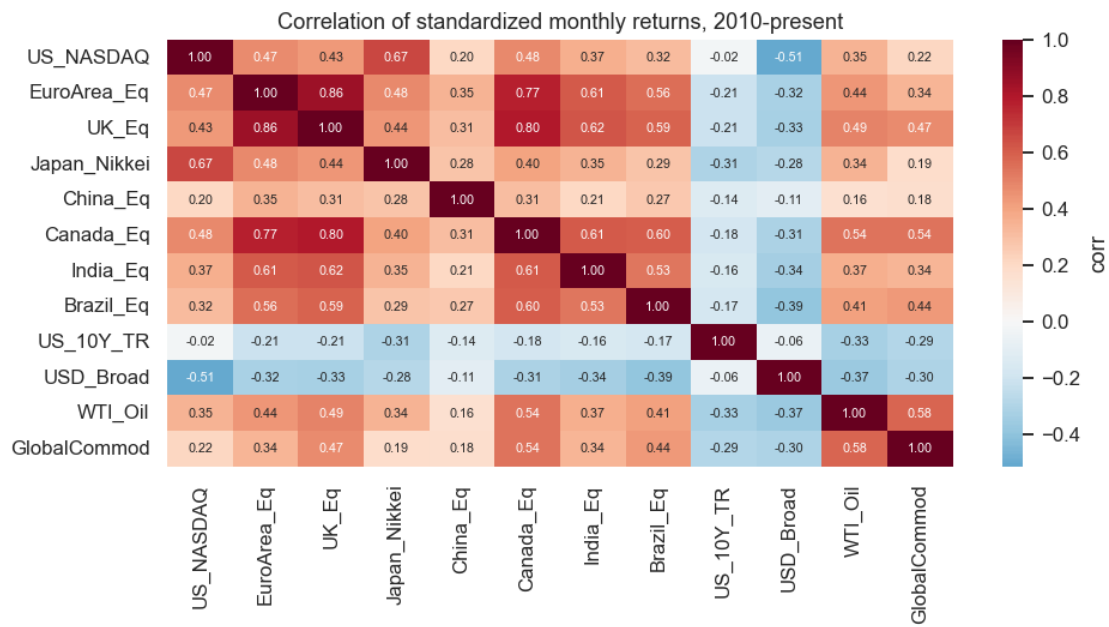
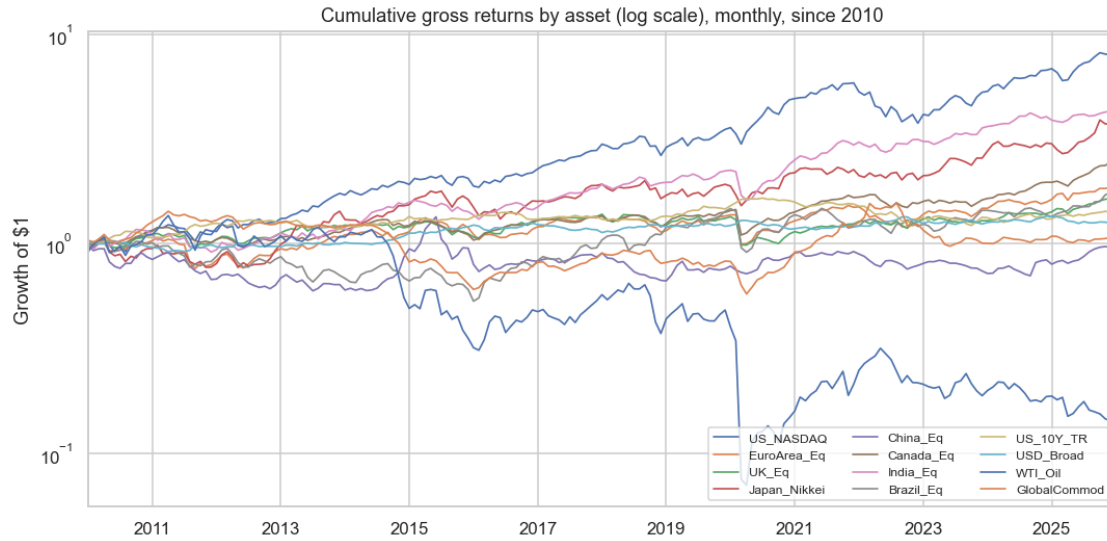
	GlobalCommod
2025-01-31	0.604152
2025-02-28	-0.480359
2025-03-31	-0.805322
2025-04-30	0.227308
2025-05-31	-0.747691

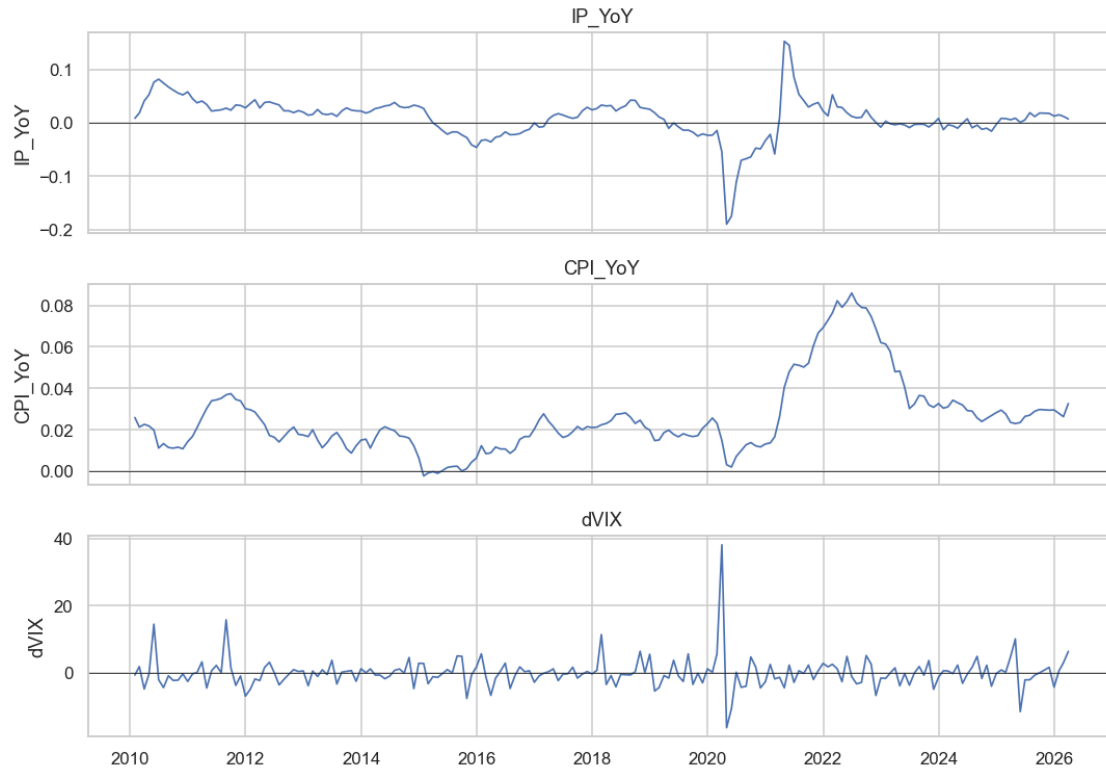
OOS PC lead/lag summary + in-sample R^2 | DataFrame | shape=(12, 4)

	cum_actual_std	cum_PCpred_std	cum_resid_std	in_sample_R2
Canada_Eq	8.408542	6.411635	1.996907	0.772665
China_Eq	3.897737	2.110845	1.786891	0.385934
Brazil_Eq	5.845632	4.790080	1.055552	0.579023

US_10Y_TR	3.663634	2.918989	0.744645	0.753834
UK_Eq	6.354611	5.804534	0.550077	0.759559

1.2 Charts





In-sample factor loadings (blue = $|t| \geq 2$ with HAC SE)

